

Unit 104 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 111.5 ac. 59 plots, BAF 20, variable-radius cruise on a 300 ft grid, 611 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	207 sq ft/ac (1 SE of mean, BA 2.0 %)	Type	Sierran mixed conifer
Trees per acre	165	CWHR size / density	4 / D (cover 85 %)
QMD	15.2 in	SDI	300 (52 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	186 sq ft/ac 18.2 in	BA QMD	21 sq ft/ac 7.8 in
TPA	103	TPA	62
Volume	24.2 MBF/ac (4817 cu ft)	Volume	7.4 green tons/ac (324 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 138 sq ft/ac BA	Vol. sampling error	9.6 % at 95 % (CV 37 %, t = 2.00, n = 59)
Species (BA)	PP 66, WF 61, IC 32, DF 29, SP 20	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 7
		Sale volume share	5.4 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.9 % at 95 % (CV 15 %); reference, not the standard
		Measured trees	PP 195, WF 179, DF 86, SP 58, IC 93; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	59; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	21	31.5	7.1	116	0.6
8-11.9	79	50.4	26.8	411	2.1
12-15.9	108	34.9	36.6	682	3.4
16-19.9	121	23.9	41.0	943	4.7
20-23.9	81	10.5	27.5	712	3.6
24-27.9	78	7.4	26.4	758	3.8
28-31.9	38	2.7	12.9	427	2.1
32-35.9	30	1.6	10.2	364	1.8
36-39.9	24	1.1	8.1	305	1.5
40-43.9	12	0.4	4.1	161	0.8
44-47.9	7	0.2	2.4	99	0.5
48-51.9	6	0.2	2.0	82	0.4
52-55.9	2	0.0	0.7	26	0.1
56-59.9	2	0.0	0.7	27	0.1
60-63.9	2	0.0	0.7	29	0.1
All	611	164.8	207.1	5141	25.8

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	195	49.2	66.1	15.7	1799	9.0
WF	179	51.6	60.7	14.7	1527	7.7
IC	93	24.7	31.5	15.3	654	3.3
DF	86	20.6	29.2	16.1	702	3.5
SP	58	18.5	19.7	14.0	459	2.3
All	611	164.8	207.1	15.2	5141	25.8

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 104 plots, 300 ft grid (orange: QA flag)

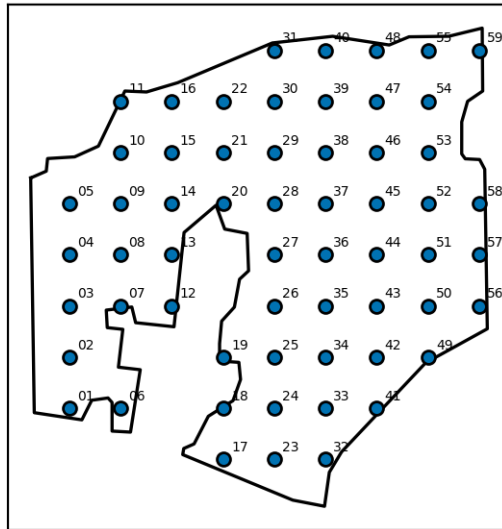


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-01	8	160	20	ok
104-02	12	240	10	ok
104-03	12	240	7	ok
104-04	11	220	9	ok
104-05	6	120	7	ok
104-06	10	200	24	ok
104-07	8	160	15	ok
104-08	11	220	9	ok
104-09	8	160	5	ok
104-10	9	180	9	ok
104-11	9	180	30	ok
104-12	10	200	33	ok
104-13	11	220	21	ok
104-14	10	200	10	ok
104-15	9	180	5	ok
104-16	8	160	10	ok
104-17	11	220	15	ok
104-18	12	240	20	ok
104-19	9	180	17	ok
104-20	12	240	33	ok
104-21	9	180	9	ok
104-22	11	220	8	ok
104-23	11	220	24	ok
104-24	10	200	18	ok
104-25	12	240	27	ok
104-26	10	200	32	ok
104-27	13	260	21	ok
104-28	11	220	13	ok
104-29	8	160	12	ok
104-30	11	220	6	ok
104-31	11	220	22	ok
104-32	11	220	13	ok
104-33	11	220	11	ok
104-34	8	160	14	ok
104-35	12	240	15	ok
104-36	13	260	13	ok
104-37	11	220	35	ok
104-38	11	220	17	ok
104-39	10	200	7	ok
104-40	11	220	10	ok
104-41	11	220	22	ok
104-42	12	240	9	ok
104-43	10	200	13	ok
104-44	12	240	6	ok
104-45	11	220	13	ok
104-46	9	180	34	ok
104-47	12	240	7	ok
104-48	9	180	22	ok
104-49	10	200	29	ok
104-50	6	120	11	ok
104-51	12	240	5	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-52	12	240	4	ok
104-53	11	220	16	ok
104-54	10	200	11	ok
104-55	12	240	4	ok
104-56	11	220	13	ok
104-57	11	220	5	ok
104-58	10	200	12	ok
104-59	9	180	2	ok

SIMULATED DATA